The purpose of the ON-OFF controller is when the voltage output from the inverting op-amp is larger than the High Threshold Voltage (V_H), the LED will turn on, and when it is lower than the Low Threshold Voltage (V_L), the LED will turn off. As such, the controller can be used in numerous applications such as thermostats, light sensors, and other switching regulators.

The operation of this ON-OFF controller can be classified into three different parts. From the left, the first op-amp is the summing amplifier, the second op-amp is the inverter, and the third one is the comparator.

The first op-amp is a summing amplifier, which serves the purpose of creating a combined signal from the input and output feedback. This allows the output to generate a certain influence on the decision boundary at the third op-amp to create hysteresis. Hysteresis refers to the phenomenon where the response output is not solely dependent on the current input of the system but also on past inputs. By introducing hysteresis into the ON-OFF controller, we eliminate the possibility of rapid switching when Vin is near the threshold by fine-tuning the comparator's threshold.

The second op-amp, the inverter, serves the purpose of inverting the polarity of the combined signal. Since the summing amplifier inverts the input signal from a sin(x) wave to a sin(x + 180°) wave, the inverter acts as an inverting buffer to ensure that the comparator receives a suitable logic flow and does not experience any unwanted phase shift. In short, it acts as a conditioner to restore the waveform to the same type as the original input.

The third op-amp acts as the comparator. The comparator is the true ON-OFF switch in the circuit. It compares the current voltage with the reference voltage (V_{ref}). If the current voltage input is larger than V_{ref}, the circuit will turn on and a voltage of 5V will be presented as the output — hence, the ON state. On the contrary, when the input voltage is lower than the Low Threshold Voltage (V_L), the output will be 0V — representing the OFF state of the ON-OFF controller.

The V_L can be obtained via the following formula:

$$V_L = \frac{R}{R_f} - V_0$$

Where V_0 is the initial voltage (5V), R is the fixed resistor (3.3 kΩ), as stated in the assignment requirements), and R_f is the resistance of the potentiometer R2.

By doing so, the entire function of the ON-OFF controller can be completed.